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(21218) **Roll No. .....**

BCA-I Sem.

18005

B. C. A. Examination, Dec. 2018

MATHEMATICS-I

(BCA-101)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required. 3×5=15

1. Show that $A = \begin{bmatrix} 3 & 1+2i \\ 1-2i & 2 \end{bmatrix}$ is Hermitian.

2. Define limit of a function at a point.

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3. Find the asymptotes of the curve $\frac{a^2}{x^2} - \frac{b^2}{y^2} = 1$.

4. State fundamental theorem of calculus.

5. Define vector in 2-dimension with example.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. Determine the values of a, b, c for which the function :

$$f(x) = \begin{cases} \dfrac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \dfrac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$$

is continuous at $x = 0$.

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7. If $y = a\cos(\log x) + b\sin(\log x)$, show that :

$x^2y_2 + xy_1 + y = 0$ and $x^2y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$.

8. Calculate the area of parallelogram spanned by the vectors $a = (1, -1, 3)$ and $b = (2, -7, 1)$.

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. Verify Cayley -Hamilton theorem for the matrix :

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Also determine the characteristic roots and corresponding characteristic vector of the matrix A.

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10. Check the continuity of the following functions at $x = 0$:

(i) $f(x) = \dfrac{|x|}{x}$ for $x \neq 0$ and $f(0) = 0$

(ii) $f(x) = \dfrac{e^{1/x}\sin(1/x)}{1+e^{1/x}}$ for $x \neq 0$ and $f(0) = 0$.

11. Trace the curve $9ay^2 = (x-2a)(x-5a)^2$.

12. (i) Expand $\dfrac{e^x}{e^x+1}$ by Maclaurin's theorem.

(ii) Expand $\log x$ in power of $(x-1)$ by Taylor's theorem.

13. Evaluate :

(i) $\int \tan^4 x\, dx$

(ii) $\int \dfrac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha}\, dx$.

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A **Printed Pages : 4**

(21119) **Roll No. .....**

BCA-I Sem.

18005

B.C.A. Examination, November 2019

MATHEMATICS-I

(BCA-101)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** Sections as per instructions.

Section-A

Note : *Attempt all the **five** question of this section. Each question carries 3 marks. Veryshort answer is required.* 5×3=15

1. Define rank of a matrix.

2. Show that $\lim_{x \to 2} \frac{|x-2|}{x-2}$ does not exist.

3. Verify Rolle's theorem for the function.

$$f(x) = 2x^3 + x^2 - 4x - 2, \ x \in \left[-\sqrt{2}, \sqrt{2}\right]$$

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4. Evaluate :

$$\int x^2 e^x \, dx$$

5. Write the formula of $\vec{a} \cdot \vec{b}$ and $\vec{a} \times \vec{b}$.

Section-B

Note : *Attempt any **two** questions out of the following three questions. Each question carries 7½ marks. Short answer is required.* 2×7½=15

6. Solve the following system of equations by Cramers Rule

$$3x + 4y = 5$$
$$x - y = -3$$

7. Differentiate $(\sin x)^x$

8. Evaluate :

$$\int \frac{xe^x}{(1+x)^2} \, dx$$

Section-C

Note : *Answer any **three** questions out of the following five questions. Each question carries 15 marks. Answer is required in detail.* 3×15=45

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9. (i) Given :

$$\bar{a} = \hat{i} + 3\hat{j} - 2\hat{k}$$

$$\bar{b} = -\hat{i} + 3\hat{k}$$

Find $\bar{a} \cdot \bar{b}$ and $|\bar{a} \times \bar{b}|$

(ii) Find the unit vector perpendicular to both the vectors

$4\hat{i} - \hat{j} + 3\hat{k}$ and $-2\hat{i} + \hat{j} - 2\hat{k}$

10. Evaluate the following Integral of limit of sum $\int_a^b x\,dx$.

11. Evaluate by L' Hospital rule

(i) $\lim\limits_{x \to 1} \dfrac{x^5 - 2x^3 - 4x^2 + 9x - 4}{x^4 - 2x^3 + 2x - 1}$

(ii) $\lim\limits_{x \to 0} \dfrac{x - \sin x}{x^3}$

12. (i) Evaluate $\lim\limits_{x \to 0} \dfrac{x - |x|}{x}$

(ii) Given $f(x) = \dfrac{|x|}{x}$, for $x \neq 0$

and $f(0) = 0$

show that $f(x)$ is not continuous at $x = 0$

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13. (i) Find the Rank of the matirx

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

(ii) Find the adjoint of the matirx

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

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(20321) Roll No. .....

BCA-I Sem.

18005

B.C.A. Examination, Dec.-2020

MATHEMATICS-I

(BCA-101)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questi sections as per instructions.

Section-A

Note : Attempt all the **five** questions of this section. Each question carries 3 marks. 5×3=15

1. Define rank of a Matrix with example.
2. Find third differential coefficient of $x^4.e^{2x}$.
3. What do you mean by Beta and Gamma function?

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4. Give the statement of Rolle's theorem.
5. In short, explain Dot product and Cross product.

Section-B

Note : Attempt any **two** questions out of the three questions. Each question carries 7½ marks. 2×7½=15

6. Solve the following equations by Cramer's Rule

$3x+4y=5$

$x-y=-3$

7. Use Maclaurin's theorem to prove that

$$\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots\ldots(-1)^{n/2}\frac{x^n}{n!}+\ldots.$$

8. If $I_n=\int_0^{\pi/3}\tan^n x\,dx$ then show that $(n-1)(I_n+I_{n-2})=(\sqrt{3})^{n-1}$

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Section-C

Note : Attempt any **three** questions out of the following five questions. Each question carries 15 marks. 3×15=45

9. What do you mean by L-Hospital rule? Evaluate $\lim_{x\to\frac{\pi}{2}} \frac{\log\left(x-\frac{\pi}{2}\right)}{\tan x}$ by using L-Hospital Rule.

10. Examine the function f(x) given by $f(x) = 10x^6-24x^5+15x^4-40x^3+108$ for maximum and minimum values.

11. If $\vec{F} = (x^2+y^2)\hat{i}-2xy\hat{j}$ and curve C is the rectangle in xy-plane bounded by y=0, x=a, y=b, x=0 then prove that $\int_C \vec{F}.d\vec{r} = -2ab^2$

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12. If $f(x)=\frac{|x|}{x}$, for $x \neq 0$

and f(x)= 0, for x=0

then show that f(x) is not continuous at x=0.

13. Investigate for what values of λ, μ the simultaneous equations

x+y+z=6

x+2y+3z=10

x+2y+λz=μ

Prove (i) no solution (ii) a unique solution and (iii) infinitely many solutions.

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A (Printed Pages 4)

(20222) Roll No. .....

BCA-I Sem.

18005 (CV-III)

B.C.A. Examination, Dec.-2021

MATHEMATICS-I

(BCA-101)

Time : 1½ Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section-A

(Very Short Answer Questions)

Note : Attempt any **two** questions of this Section. Each question carries **7.5** marks. Very short answer is required. 2×7.5=15

1. Define continuity at a point.
2. State Caley-Hamilton Theorem.

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3. If $y = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$, find $\frac{dy}{dx}$
4. Evaluate $\int \log_e x \, dx$.
5. Find λ such that $\bar{a}$ and $\bar{b}$ are perpendicular vector where

   $\bar{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$; $\bar{b} = \hat{i} - 2\hat{j} + 3\hat{k}$.

Section-B

(Short Answer Questions)

Note : Attempt any **one** question out of the following **three** questions. Each question carries **15** marks. Short answer is required. 1×15=15

6. Expand e^x in ascending powers of x by Maclaurin's theorem.
7. Differentiate x^x.
8. Prove that

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c).$$

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Section-C

(Detailed Answer Questions)

Note : Attempt any **two** questions out of the following **five** questions. Each question carries **22.5** marks. Answer is required in detail. 2×22.5=45

9. (a) Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

(b) Determine the eigen values of the matrix

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

10. (a) Show that the function $f(x)=|x|$ is continuous at $x=0$.

(b) Evaluate

$$\lim_{h \to 0}\left(\frac{\sqrt{x+h}-\sqrt{x}}{h}\right)$$

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11. (a) Differentiate $y=x \sin x \log x$.

(b) Find the maximum and minimum values of $(3x^4-8x^3+12x^2-48x+25)$ in $[0, 3]$.

12. (a) Evaluate $\int x^2 \sin x \, dx$.

(b) Evaluate $\int (\sqrt{\sin x} \cdot \cos x) dx$.

13. (a) Show that the vectors $\hat{i}-3\hat{j}+4\hat{k}, 2\hat{i}-\hat{j}+2\hat{k}$ and $4\hat{i}-7\hat{j}+10\hat{k}$ are coplanar.

(b) Find the area of a parallelogram whose adjacent sides are determined by the vectors.

$\vec{a} = \hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b} = -3\hat{i}-2\hat{j}+\hat{k}$.

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(21121) (Printed Pages 4)

B.C.A.-I Sem. Roll No. .....

18005 (CV-II)

B.C.A. Spl. and Back Paper Examination, Nov.-2021

MATHEMATICS-I

(BCA-101)

Time : 1½ Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section-A

Note : Attempt any **two** questions of this section. Each question carries 7.5 marks. Short answer is required.

2×7.5=15

1. Define Eigen value and Eigen Vector of a matrix.

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2. Evaluate $\lim_{x \to 0}\left(\frac{1}{x}\right)$.

3. Explain Beta function and Gamma function.

4. Find the second differential coefficient of $x^4.e^{5x}$.

5. What is the difference between Scalars and Vectors. Explain in brief with some example.

Section-B

Note : Attempt any **one** question out of the following three questions. Each question carries **15** marks. 1×15=15

6. Find $\lim_{x \to 1}\left(\frac{x^5 - 2x^3 - 4x^2 + 9x - 4}{x^4 - 2x^3 + 2x - 1}\right)$.

7. By using Maclaurin's theorem, prove that $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\ldots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \ldots\ldots$

8. If $u_n = \int x^n.(a-x)^{3/2}\, dx$ then show that $(2n+3)\, u_n = 2an\, u_{n-1} - 2x^n(a-x)^{3/2}$.

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Section-C

Note : Attempt any **two** questions out of the following five questions. Each question carries **22.5** marks.

$2 \times 22.5 = 45$

9. Check the continuity and differentiability of the function defined by $f(x)=|x|$ at $x=0$.

10. Explain Cramer's Rule. Solve the following equations

$2x-y+3z=9$

$x+y+z=6$

$x-y+z=2$ by Cramer's Rule.

11. If $\vec{r}(t) = 5t^2\hat{i} + t\hat{j} - t^3\hat{k}$, then

Prove that

$$\int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2}\right) dt = -14\hat{i} + 75\hat{j} - 15\hat{k}.$$

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12. If $I_n = \int_0^{\pi/4} \tan^n x \, dx$ then prove that

$$I_n + I_{n-2} = \frac{1}{n-1}$$

and deduce the value of I_5.

13. Verify Rolle's theorem for the function $f(x)=(x^2+2x-3)e^x$ in the interval $[-3, 1]$.

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11. Verify Cayley-Hamilton theorem of the matrix A' and hence find A^{-1} where

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

12. If $\hat{a}$ and $\hat{b}$ are Unit Vectors inclined at an angle 'A', then prove that

(i) $\sin\left(\frac{A}{2}\right) = \frac{1}{2}|a - \hat{b}|$

(ii) $\cos\left(\frac{A}{2}\right) = \frac{1}{2}|\hat{a} + \hat{b}|$

(iii) $\tan\left(\frac{A}{2}\right) = \frac{|\hat{a} - \hat{b}|}{|\hat{a} + \hat{b}|}$

13. Evaluate the following integrals :

(a) $\int x^2 \sin x \, dx$

(b) $\int x\sqrt{x+2}\, dx$

(c) $\int \frac{x}{(x^2+2)(x^2+3)} dx$

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A (Printed Pages 4)

(21222) Roll No. .....

BCA-I Sem.

18005

B.C.A. Examination, Dec.-2022

MATHEMATICS - I

(BCA - 101)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section - A

(Very Short Answer Type Questions)

Note : Attempt **all** questions of this section. Each question carries 3 marks.

3×5=15

1. Give an example of matrices A, B such that AB=0 but $A \neq 0, B \neq 0$.

2. Verify Rolle's theorem for the function $f(x)=\sqrt{4-x^2}$ in the interval [−2, 2].

3. Evaluate $\lim_{x\to 0} \frac{\tan x - x}{x^2 \tan x}$.

4. Write the relationship between Gamma and Beta function.

5. Find the characteristic roots of the matrix

$$A=\begin{bmatrix} a & h & g \\ 0 & b & f \\ 0 & 0 & c \end{bmatrix}$$

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions out of the following three questions. Each question carries 7½ marks.

7½×2=15

6. Explain $\log_e(1+x)$ in ascending powers of 'x'.

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7. Solve by Cramer's Rule :

$$5x-7y+z=11$$
$$6x-8y-z=15$$
$$3x+2y-6z=7$$

8. Prove that Conical text of given capacity will require the least amount of Canvas when the height is $\sqrt{2}$ times the radius of the base.

Section-C

(Long Answer Type Questions)

Note : Attempt any **three** questions out of the following five questions. Each question carries 15 marks. 3×15=45

9. Trace the curve

$$4ay^2 = x\,(x-2a)^2$$

10. If $y=(\sin^{-1}x)^2$ prove that $(1-x^2)y_2 - xy_1 - 2 = 0$ also prove that $(1-x^2)y_{n+2} - x(2n+1)y_{n+1} - n^2y_n = 0$

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(21223) Roll No. .....

B.C.A.-I Sem.

18005

B.C.A. Examination, Dec.-2023

MATHEMATICS-I

(BCA-101)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** the Sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt **all** questions of this section. Each question carries 3 marks.

$3\times5=15$

1. Find the rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 4 & 6 \end{bmatrix}$

2. Evaluate $\lim_{x \to 0} \frac{\sin 4x}{x}$

3. Differentiate xe^x function w.r. to x.

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4. **Evaluate** $\int \frac{1}{\sqrt{4x+3}}\, dx$

5. If $\vec{a} = 2\hat{i} + 4\hat{j} + 7\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} - 5\hat{k}$ Find $\vec{a}.\vec{b}$.

Section-B

(Short Answer Type Questions)

Note : Attempt any two questions out of the following three questions. Each question carries 7½ marks.

$7.5\times2=15$

6. Expand e^x in ascending powers of x upto four terms.

7. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ show that $A^{-1} = \frac{1}{19}A$

8. Show that $f(x)=|x|$ is continuous at $x=0$.

Section-C

(Long Answer Types Questions)

Note : Attempt any three questions out of the following five questions. Each question carries 15 marks. $3\times15=45$

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9. Obtain the characteristic equation of the matrix $A = \begin{vmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{vmatrix}$ and verify that it is satisfied by A.

10. Find :

(a) $D^n(x^3e^x)$

(b) If $y=(\sin^{-1}x)^2$ prove that

$(1-x^2)y_2-xy_1-2=0$

11. Find 'C' of the Lagrange's mean value theorem for the function $f(x)=2x^2-10x+29$ in [2, 7].

12. Evaluate :

(a) $\int \frac{dx}{x(x^3+1)}$.

(b) $\int \frac{dx}{(x+1)(x+2)}$

(c) $\int (x^2+1)e^x\,dx$

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13. If $|\vec{a}| = 2$, $|\vec{b}| = 7$ and $\vec{a}\times\vec{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, Find the angle between vectors $\vec{a}$ and $\vec{b}$.

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